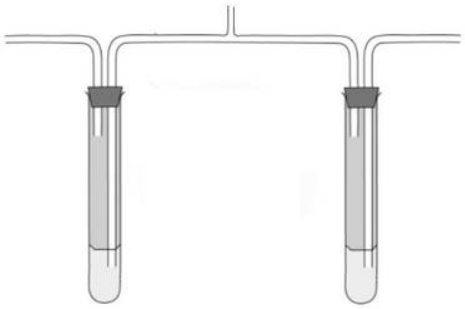
# Type MS · Explain how respiration releases energy and ATP transfers it

## Answer 1 of 7 · Question 7 part b · 5 marks

2024 JUN · Paper 02 (regional R) · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 7 (a) | (i) | glucose ⟶ lactic acid/lactate; | ignore any ref to $CO_2$ and $H_2O$ on right side of equation | (1) |
| | (ii) | $C_6H_{12}O_6 + 6O_2; \longrightarrow 6H_2O + 6CO_2;$ | | (2) |
| | (iii) | general diagram;<br>tubing correct way round;<br>  | labels not needed<br><br>long short, long short<br>short long, short long | (2) |
| | (iv) | • inhale and exhale/breathe through the mouthpiece;<br>• at rest;<br>• time how long it takes the limewater to go cloudy/milky;<br>• repeat following exercise; | | (4) |
| (b) | | any five from<br><br>• energy released from (breakdown of) glucose;<br>• during (aerobic) respiration;<br>• (energy used) to add phosphate group;<br>• to convert ADP to form ATP;<br>• energy is released from ATP breakdown of ATP releases energy;<br>• breaking of phosphate bond/ATP hydrolysed/ATP breaks down into ADP + P(i);<br>• to provide energy for muscle contraction; | | (5) |

**Total for Question 7 = 14 marks**

# Type MS · Explain how respiration releases energy and ATP transfers it

## Answer 2 of 7 · Question 3 part b.iii · 2 marks

2020 JAN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (i) | **Structure** — **Function**<br>cell membrane → helps the cell to move<br>ribosomes → make proteins<br>flagellum → holds the genetic code<br>controls which substances enter and exit the cell → protection against pathogens | Reject more than one line from each structure<br><br>One mark for each correct line | 3 |
| (ii) | Any two from:<br>• (bacterial cell) has a cell wall (1)<br>• (bacterial cell) does not have a nucleus (1)<br>• (bacterial cell) has a plasmid (1) | | 2 |
| (b) (i) | **The only correct answer is B diffusion**<br><br>*A is incorrect because it is down a diffusion gradient*<br><br>*C is incorrect because it is not removal from the bacterium*<br><br>*D is incorrect because osmosis only involves movement of water* | | 1 |
| (ii) | • glucose (1)<br>• carbon dioxide (1) | Answers must be in the correct order<br>Allow correct chemical formulae | 2 |
| (iii) | Any two from:<br>• to provide energy (1)<br>• for movement (1)<br>• for growth/reproduction/divide/mitosis (1)<br>• for chemical reactions/producing organelles/named organelle (1) | | 2 |

## Type MS · Explain how respiration releases energy and ATP transfers it

### Answer 2 of 7 · Question 3 part b.iii · 2 marks

2020 JAN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

**Total for Question 3 = 10 marks**

## Type MS · Explain how respiration releases energy and ATP transfers it

### Answer 3 of 7 · Question 7 part a.i · 1 mark

2020 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 7 (a) (i) | | aerobic respiration (1) | | 1 |
| | (ii) | • shorter (1)<br>• fatter/wider/thicker (1) | | 2 |
| | (iii) | • shortening causes tension/pull (1)<br>• tendon inelastic (1)<br>• pulls on radius/bone (1) | | 3 |
| (b) | | • ball and socket indicated labelled bone (1)<br>• capsule/ligament labelled (1)<br>• cartilage labelled (1)<br>• synovial fluid labelled (1)<br>• synovial membrane labelled (1)<br>• overall appearance of diagram (1) | | 6 |

Total 12 marks

## Type MS · Explain how respiration releases energy and ATP transfers it

### Answer 3 of 7 · Question 7 part a.i · 1 mark

2020 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

## Type MS · Explain how respiration releases energy and ATP transfers it

### Answer 4 of 7 · Question 5 part a.ii · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 5 | (a) | (i) | $C_6H_{12}O_6$;<br>$6H_2O$; | | 2 |
| | | (ii) | ATP breaks down/hydrolysed;<br>to ADP and Pi; | Allow phosphate/P | 2 |
| | (b) | (i) | • suitable axis labels with units;<br>• time on x axis;<br>• suitable scale;<br>• correct plots; | | 4 |
| | | (ii) | • suitable curve; | | 1 |
| | | (iii) | Five from;<br>• mass of haemoglobin increases (with time/training at altitude);<br>• increased number of red blood cells/more haemoglobin in the blood;<br>• haemoglobin binds oxygen/red blood cells carry oxygen;<br>• more oxygen transported around the body/in the blood;<br>• to muscle (cells/tissue);<br>• for aerobic respiration;<br>• to release energy; | | 5 |

Total 14 marks

## Type MS · Explain how respiration releases energy and ATP transfers it

### Answer 4 of 7 · Question 5 part a.ii · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Explain how respiration releases energy and ATP transfers it

### Answer 5 of 7 · Question 4 part a.iv · 1 mark

2023 JUN · Paper 02 · extract 1 of 2 · find part a.iv in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 4 | (a) | (i) | • glucose;<br>• carbon dioxide and water; | Reject formulae | 2 |
| | | (ii) | aerobic; | | 1 |
| | | (iii) | B; (mitochondrion) | | 1 |
| | | (iv) | ATP/adenosine triphosphate | | 1 |
| | (b) | (i) | Any two from<br>• more people/repeat tests:<br>• different gender/diet/location/age;<br>• carry out test over more years/longer time period;<br>• calculate the mean/ average; | | 2 |
| | | (ii) | • 7250 - 6550; | ecf if wrong figures used in a subtraction calculation but correct answer 1 mark | |
| | | | • = 700kJ; | | 2 |
| | | (iii) | Any five from<br>• metabolic rate increases in colder temperatures;<br>• correct comparative reading from graph e.g. metabolic rate is higher in January compared to July;<br>• (in colder temperatures) increased rate of respiration;<br>• releases heat (energy)/more energy needed;<br>• (energy needed) to maintain body temperature/reference to keeping body warm;<br>• (in warmer weather) more heat lost (by radiation / convection/sweating); | ORA for mps 1, 3, 4, 5, 6<br>Allow marking point if correct figures given from graph | 5 |
| | | (iv) | uptake of oxygen/carbon dioxide output; | Allow amount of glucose used | 1 |

Total 15

## Type MS · Explain how respiration releases energy and ATP transfers it

### Answer 5 of 7 · Question 4 part a.iv · 1 mark

2023 JUN · Paper 02 · extract 2 of 2 · find part a.iv in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Explain how respiration releases energy and ATP transfers it

### Answer 6 of 7 · Question 4 part b.iii · 5 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 4 | (a) | (i) | • glucose;<br>• carbon dioxide and water; | Reject formulae | 2 |
| | | (ii) | aerobic; | | 1 |
| | | (iii) | B; (mitochondrion) | | 1 |
| | | (iv) | ATP/adenosine triphosphate | | 1 |
| | (b) | (i) | Any two from<br>• more people/repeat tests:<br>• different gender/diet/location/age;<br>• carry out test over more years/longer time period;<br>• calculate the mean/ average; | | 2 |
| | | (ii) | • 7250 - 6550; | ecf if wrong figures used in a subtraction calculation but correct answer 1 mark | |
| | | | • = 700kJ; | | 2 |
| | | (iii) | Any five from<br>• metabolic rate increases in colder temperatures;<br>• correct comparative reading from graph e.g. metabolic rate is higher in January compared to July;<br>• (in colder temperatures) increased rate of respiration;<br>• releases heat (energy)/more energy needed;<br>• (energy needed) to maintain body temperature/reference to keeping body warm;<br>• (in warmer weather) more heat lost (by radiation / convection/sweating); | ORA for mps 1, 3, 4, 5, 6<br>Allow marking point if correct figures given from graph | 5 |
| | | (iv) | uptake of oxygen/carbon dioxide output; | Allow amount of glucose used | 1 |

Total 15

## Type MS · Explain how respiration releases energy and ATP transfers it

### Answer 6 of 7 · Question 4 part b.iii · 5 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Explain how respiration releases energy and ATP transfers it

### Answer 7 of 7 · Question 4 part b.iv · 1 mark

2023 JUN · Paper 02 · extract 1 of 2 · find part b.iv in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 4 | (a) | (i) | • glucose;<br>• carbon dioxide and water; | Reject formulae | 2 |
| | | (ii) | aerobic; | | 1 |
| | | (iii) | B; (mitochondrion) | | 1 |
| | | (iv) | ATP/adenosine triphosphate | | 1 |
| | (b) | (i) | Any two from<br>• more people/repeat tests:<br>• different gender/diet/location/age;<br>• carry out test over more years/longer time period;<br>• calculate the mean/ average; | | 2 |
| | | (ii) | • 7250 - 6550; | ecf if wrong figures used in a subtraction calculation but correct answer 1 mark | |
| | | | • = 700kJ; | | 2 |
| | | (iii) | Any five from<br>• metabolic rate increases in colder temperatures;<br>• correct comparative reading from graph e.g. metabolic rate is higher in January compared to July;<br>• (in colder temperatures) increased rate of respiration;<br>• releases heat (energy)/more energy needed;<br>• (energy needed) to maintain body temperature/reference to keeping body warm;<br>• (in warmer weather) more heat lost (by radiation / convection/sweating); | ORA for mps 1, 3, 4, 5, 6<br>Allow marking point if correct figures given from graph | 5 |
| | | (iv) | uptake of oxygen/carbon dioxide output; | Allow amount of glucose used | 1 |

Total 15

## Type MS · Explain how respiration releases energy and ATP transfers it

### Answer 7 of 7 · Question 4 part b.iv · 1 mark

2023 JUN · Paper 02 · extract 2 of 2 · find part b.iv in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|